

PROFILE

Computer Science student targeting C/C++ systems, graphics, engine tooling, and low-level software roles, with projects in engine architecture, CPU rendering, interpreters, and digital logic.

EDUCATION

Newcastle University	Newcastle, UK
BSc Computer Science; 50% Vice-Chancellor’s Excellence Scholarship — First Class Honours	Expected 2028
METU Development Foundation High School	Turkey
GPA: 84.3/100 · IELTS: 7.5	Graduated 2025

SKILLS

Languages: C, C++, Rust, Lua, Assembly

Graphics/Engine: OpenGL, GLSL, SDL2/3, OpenAL, Box2D, Unity

Tools: Git, CMake, CLion, JSON, Logisim Evolution

Spoken: Turkish, English, Arabic

EXPERIENCE

AdresGezgini	
Software/Game Development Intern	Jul 2024 – Aug 2024
o Built a 2D game prototype and supported client website debugging by reproducing issues, testing fixes, and identifying front-end problems.	
London Success Academy	
Software Testing Intern	Mar 2025 – Apr 2025
o Supported QA workflows by reproducing defects, documenting issues, and testing software behaviour against expected outcomes.	

PROJECTS

- Tilky Engine — Open-Source Game Engine and Editor C++20/Rust, OpenGL, SDL3, ImGui, Lua, OpenAL, Tracy
- o Built and maintained a complete C++20 engine toolchain with launcher, runtime, visual editor, asset/project pipeline, scripting, audio, rendering, localisation, profiling, and standalone export support.
- o Designed a sector-based 3D rendering architecture inspired by classic Doom/Build-style engines, using real 3D quad primitives, textured walls/floors, decals, billboards, depth buffering, and SSBO-backed geometry buffers.
- o Implemented a data-oriented ECS with entity IDs and type-specific component arrays, supporting cache-friendly updates, efficient runtime iteration, and sector-aware collision and physics systems.
- o Developed an ImGui-based level editor with wall/sector creation, texture selection, entity/component editing, save/load workflows, audio world settings, and fast editor-to-runtime testing.
- o Optimized collision using dirty flags and current/neighbor-sector filtering, profiling a median collision pass of approximately 44.97 microseconds with 13 entities in the same sector.
- o Added Lua scripting, OpenAL positional 3D audio, Rust-based standalone exporting, and Rust localisation tooling that generates translator-ready JSON files without requiring technical setup.
- C Software Renderer — CPU-Side Textured 3D Renderer
- o Built a scratch-implemented CPU rasterisation pipeline with custom math structures for world-to-camera transforms, clipping, perspective projection, and depth buffering.
- o Implemented textured triangle/quad rendering with UV interpolation and texture sampling, profiling close-up scenes to identify CPU-side rasterisation bottlenecks.
- C Interpreter — Tree-Walking Language Implementation
- o Developed a custom tree-walking interpreter with a hand-written lexer, recursive-descent parser, AST representation, variables, expressions, conditionals, scoped blocks, and return values.
- o Implemented user-defined functions, recursive function calls, and runtime scope handling while manually managing AST/runtime memory in C.
- Custom 8-Bit CPU Architecture — Digital Logic Design
- o Designed an 8-bit CPU from digital logic components, implementing an ALU, registers, program counter, instruction decoder, and control logic.
- o Defined a custom instruction set and tested execution flow by simulating simple assembly-style programs in Logisim Evolution.

ACTIVITIES

Robotics team lead for a 5-member international competition team; led multiple 3–7 day game jam projects; co-produced and performed a paid stage play invited to an international festival.